

Variable reaction coordinate transition state theory: Analytic results and application to the $\text{C}_2\text{H}_3 + \text{H} \rightarrow \text{C}_2\text{H}_4$ reaction

Yuri Georgievskii and Stephen J. Klippenstein

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Variable reaction coordinate transition state theory: Analytic results and application to the $\text{C}_2\text{H}_3 + \text{H} \rightarrow \text{C}_2\text{H}_4$ reaction

Yuri Georgievskii^{a)} and Stephen J. Klippenstein

Combustion Research Facility, Sandia National Laboratories, Livermore, California 94551-0969

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A novel derivation is provided for the canonical, microcanonical, and energy E and total angular momentum J resolved reactive flux within the variable reaction coordinate transition state theory (VRC-TST) formalism. The use of an alternative representation for the reaction coordinate velocity yields a new expression for the kinematic factor which better illustrates its dependence on the pivot point location, and which can be straightforwardly evaluated. Also, the use of a geometric approach in place of an earlier algebraic one clarifies the derivation as does the use of Lagrange multiplier methodology for the analytic integration over the total angular momentum. Finally, a quaternion representation for the fragment and line-of-centers orientations is employed in place of the Euler angle or internal/external rotational coordinates used in prior studies. The result is an efficient, and particularly easy to implement, methodology for performing variable reaction coordinate transition state theory calculations. Furthermore, the simplicity of the derivation allows for the straightforward generalization to alternative forms for the dividing surface, as is illustrated by deriving the expressions for the cases of elliptical and planar dividing surfaces. Application to the $\text{C}_2\text{H}_3 + \text{H}$ reaction yields results for the total rate coefficient that are generally only 15% greater than those obtained from related trajectory simulations, thereby demonstrating the accuracy of the VRC-TST formalism. Meanwhile, results for the two separate addition channels (frontside and backside) illustrate the difficulty of accurately apportioning the total flux and particularly the inadequacy of canonical predictions for the channel specific optimized dividing surfaces. © 2003 American Institute of Physics. [DOI: 10.1063/1.1539035]

I. INTRODUCTION

Variational transition state theory (VTST) is an important conceptual and numerical tool for the study of chemical reaction kinetics.¹ The transition state dividing surface, obtained via minimization of the TST predicted reactive flux, provides a valuable approximate division of phase space into reactants and products. As such, this dividing surface serves as a natural starting point for more accurate, but generally more computationally intensive quantum and classical dynamical simulations.^{2,3} Notably, VTST predictions for the rate constant are often in good agreement with those obtained from such dynamical simulations.^{4–6}

The central focus of any application of VTST is the evaluation of the reactive flux through an arbitrary dividing surface separating the reactants from the products. At the energy E and total angular momentum J resolved level, this reactive flux, $N(E, J)$, is commonly interpreted as the number of available states (i.e., with energies less than E) with the specified total angular momentum J for motion on the dividing surface. Traditionally, the dividing surfaces have been defined in terms of a single parameter, which has been termed the reaction coordinate s . The variational minimization of the reactive flux is then performed with respect to the value of s .

For barrierless reactions, the separation between the two

reacting fragments for the optimal dividing surface varies greatly with E and/or J . Furthermore, the relative motions of the two reacting fragments can be of large amplitude. These two factors make it difficult to obtain quantitatively accurate VTST based estimates for the reactive flux. A variable reaction coordinate (VRC) implementation of TST,^{7,8} in which both the definition and the value of the reaction coordinate are optimized, has found considerable utility for such reactions. Within this scheme, the dividing surfaces are defined to have a fixed distance between individual pivot points tied to the framework of each of the reacting fragments. The variational minimization of the reactive flux is then performed with respect to both the location of the pivot points and the distance between them. This scheme allows the reaction coordinate to vary from that appropriate at large interfragment separations (e.g., the separation between the fragments' centers of mass) to that appropriate at small interfragment separations (e.g., the distance between the orbitals of the incipient bond).

Within this scheme, it has proven convenient to divide the modes into two sets termed the conserved and the transitional modes.⁹ The conserved modes correspond to the internal vibrational modes of the two reacting fragments. The character of these vibrational modes generally does not change during the reaction process. As such, their coordinates should not enter into the definition of the variational dividing surface. In contrast, the transitional modes, which include the rotations of both fragments as well as the relative

^{a)}Electronic mail: ygeorgi@mailgate.sandia.gov

translational motion of them, change from free rotation and translation at infinite separation of the fragments to vibrational motion (and overall rotation) upon formation of the bond between the fragments.

With an assumed separability of the transitional and conserved modes, $N(E, J)$ can be rewritten as⁹

$$N(E, J) = \int dE' \rho^{(c)}(E - E') N^{(t)}(E', J), \quad (1.1)$$

where $\rho^{(c)}(E)$ is the density of states for the conserved modes and $N^{(t)}(E, J)$ is the number of states for the transitional modes. The density of states for the conserved modes can be evaluated quantum mechanically using direct counting algorithms. The low frequency nature of the transitional mode motions allows one to consider classical phase space integral expressions for their contribution to the number of states. The evaluation of such phase space integrals can be computationally intensive and various studies have derived efficient means for evaluating them, focusing largely on analytic integrations over the generalized momenta.^{8,10,11}

In this paper we first derive an efficient alternative method for evaluating the transitional mode contribution to the number of states by integrating analytically over the generalized momenta. This derivation is analogous to that of Smith¹⁰ but differs in a number of key aspects. One key distinction involves the use of a new expression for the reaction coordinate velocity, which ultimately results in a simple and more physically illuminating expression for the dependence of the kinematic factor on the pivot point location. In particular, the derivatives of the reaction coordinate with respect to the angle of rotation about the principle axes, whose evaluation apparently requires either complex unspecified analytical transformations or numerical differentiation in Smith's work,¹⁰ are replaced here by simple vector products of principle axis vectors and pivot point vectors.

Other important distinctions include the use of a geometric, rather than algebraic approach, in the initial separation of coordinate and momenta terms. This approach helps to clarify the distinction between the choice of coordinates and the choice of momenta. Various other simplifications include the use of the Lagrange multiplier methodology, which provides a more direct and comprehensible procedure for integrating over the total angular momentum. Finally, focusing on absolute orientational coordinates, instead of internal and external rotational coordinates, again simplifies the implementation, while yielding no reduction in efficiency for Monte Carlo based integration routines. The net result is an expression that, while formally equivalent to earlier results,^{8,10} differs significantly in form and ease of implementation, and whose derivation is both easier to follow and to generalize. To illustrate the ease of generalizing the present result we also provide explicit expressions for the cases of elliptical and planar dividing surfaces in place of the standard spherical dividing surface.

A second aspect of this work involves the application of this methodology to a study of the $H + C_2H_3$ addition reaction. A detailed comparison between VRC-TST predictions and experiment for this reaction was recently provided.¹² Here, we focus instead on a detailed comparison of the pre-

dictions from both the E/J resolved and canonical implementations with trajectory simulations of the capture process. In doing so we consider both the total flux and the relative flux for each of the two separate addition channels.¹² The comparison with trajectory simulations provides a valuable test of the VRC-TST theoretical framework. Meanwhile, the channel specific calculations illustrate some of the difficulties that can arise in trying to apportion the total reactive flux to specific channels.

The analytical derivations are presented in Sec. II. Then, in Sec. III, the application to the reaction of vinyl radicals with hydrogen atoms is considered. Finally, some concluding remarks are provided in Sec. IV.

II. THEORY

A. Spherical dividing surface

The number of states for the transitional modes may generally be written as¹³

$$N^{(t)}(E, J) = (2\pi)^{-\nu} \int d^\nu Q d^\nu P \delta(s(Q) - s') \times \delta(H(Q, P) - E) \delta(\hat{J}(Q, P) - J) \dot{s} \Theta(\dot{s}). \quad (2.1)$$

Here, and in the following, we use regular script for dynamical variables, the calligraphic script for general, n -dimensional vectors, the bold script for three-dimensional vectors, and atomic units, in which $\hbar = 1$. The number of states $N^{(t)}$ also depends on the shape of the dividing surface defined by the reaction coordinate $s(Q)$ and on its particular value s' . This dependence is not shown explicitly but is implied. The dynamical variable for the total angular momentum is denoted as $\hat{J}$, while its numerical value is denoted as J . In Eq. (2.1) $Q = (Q_1, \dots, Q_\nu)$ and $P = (P_1, \dots, P_\nu)$ are generalized canonical coordinates and corresponding conjugate momenta, and ν is the total number of degrees of freedom for the transitional modes; excluding three modes of translational motion as a whole. The quantity $\hat{J}(Q, P)$ is the total angular momentum in the laboratory reference frame, $H(Q, P)$ is the system Hamiltonian, δ is the delta function, Θ is the Heaviside step function, and dot denotes the time derivative, $\dot{s} = ds/dt$.

The focus of the following derivation involves an analytic integration over the generalized momenta and also the reaction coordinate. Naturally, the first step in this derivation involves a choice of representation for the generalized momenta. Ordinarily, one chooses them to be conjugate to some set of coordinates, due to the invariance of the measure $d^\nu Q d^\nu P$ under canonical transformations of the generalized coordinates Q and momenta P . Here, however, we find it convenient to choose the projections $J_i^{(k)}$ of the fragment angular momentum on their principle axes for the generalized momenta representing the rotational motion of each fragment (the index $i = 1, 2, 3$ refers to a spatial axis and $k = 1, 2$ is the fragment number). Unfortunately, this representation does not allow one to make a selection of generalized coordinates that are canonically conjugate to the generalized momenta everywhere. Thus, one must consider more general, noncanonical transformations.

Choosing such a noncanonical set of generalized momenta affects the integration measure $d^{\nu}Qd^{\nu}\mathcal{P}$ in that an additional Jacobian factor w corresponding to that for the transformation in momenta appears under the integral in Eq. (2.1). Since, in effect, only linear transformations in the momentum space $\mathcal{P}$ are considered w depends only on Q and not on $\mathcal{P}$. The resultant combination $w(Q)d^{\nu}Q$ can be viewed as a measure in the configurational space which corresponds to our choice of the momentum representation. Thus, choosing the representation of the momenta induces the integration measure in the configurational space. This integration measure can be determined by realizing that locally, for small rotations, the canonically conjugate coordinate for each angular momentum component is the rotational angle around the corresponding axis. Thus, the integration measure in the space of orientations of each fragment which is induced by our choice of the momenta representation coincides with the standard invariant measure on the rotational group, which can be written using Euler angles as $\sin\vartheta d\vartheta d\varphi d\psi$. The total volume of the orientational space by this measure is $8\pi^2$. We will denote this measure symbolically as $d\Omega$, implying that it is the three-dimensional solid angle.

For linear fragments the projections of the angular momentum $J_i^{(k)}$, $i=1,2$; $k=1,2$ to two axes perpendicular to the molecular axis can be used. Similar considerations lead one to conclude that the corresponding measure induced in the orientational space of the linear fragment is simply the solid angle $d\Omega$ describing the orientation of its molecular axis, with a volume measure of 4π . Furthermore, in the present representation, the generalized coordinates and momenta related to the rotational motion of either fragment and to the relative translational motion are not intermixed. Thus, each motion can be considered separately.

For the relative translational motion of the fragments we start with the canonical coordinates given by the Cartesian coordinates of the vector $\mathbf{r}$ connecting the centers of mass of the two fragments and the corresponding conjugate momenta, $\mathbf{p}$, for the relative motion. With this representation $\mathbf{p} = \mu\mathbf{v}$, where $\mu = m_1m_2/(m_1+m_2)$ is the reduced mass, and $\mathbf{v}$ is the relative velocity of the fragments. Then, it is convenient to transform coordinates (but not momenta) from $\mathbf{r}$, to the vector $\mathbf{s}$ which connects the pivot points, cf. Fig. 1,

$$\mathbf{s} = \mathbf{r} + \mathbf{d}^{(1)} - \mathbf{d}^{(2)}, \quad (2.2)$$

where $\mathbf{d}^{(k)}$, $k=1,2$ is the vector connecting the center of mass of the k th fragment to its pivot point. Since $\mathbf{d}^{(k)}$ depends only on the orientations of the fragments (and not on $\mathbf{r}$) the Jacobian of the transformation from $\mathbf{r}$ to $\mathbf{s}$ is unity. Using spherical polar coordinates to represent $\mathbf{s}$, $d^3\mathbf{s} = s^2 ds d\Omega^{(12)}$, one can integrate out the delta function $\delta(s-s')$ in Eq. (2.1), leaving as an integration variable only the solid angle $d\Omega^{(12)}$ characterizing the direction of $\mathbf{s}$.

To summarize what was said in the last four paragraphs, Eq. (2.1) can be rewritten in the chosen representation as

$$N^{(t)}(E, J) = (2\pi)^{-\nu} s'^2 \int d^{\nu-1}Q d^{\nu}\mathcal{P} \delta(H(Q, \mathcal{P}) - E) \times \delta(\hat{\mathbf{J}}(Q, \mathcal{P}) - \mathbf{J}) \dot{s}\Theta(\dot{s}), \quad (2.3)$$

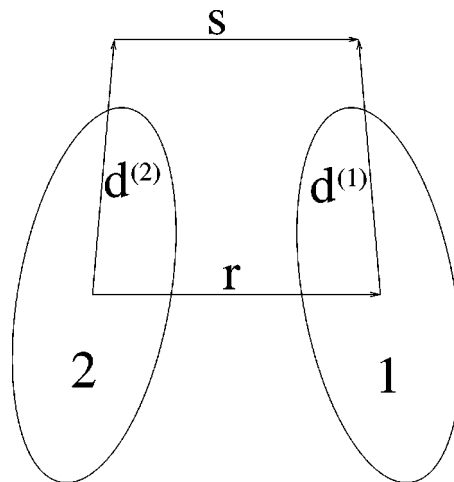


FIG. 1. Cartoon showing the definitions of the pivot point vectors, $\mathbf{d}^{(k)}$, $k=1,2$, the vector $\mathbf{r}$ connecting the centers of mass of the fragments, and the vector $\mathbf{s}$ connecting the pivot points.

where Q denotes the solid angles ($\Omega^{(k)}$ for both fragments and $\Omega^{(12)}$ for the direction of $\mathbf{s}$), $\mathcal{P}$ denotes all the momenta, (p_i , $i=1,2,3$, and $J_i^{(k)}$, $i=1,\dots,\nu_k$), $\nu = \nu_1 + \nu_2 + 3$ where ν_k is the dimensionality of the orientational space for each fragment, and the integrand is evaluated at $s=s'$.

For numerical applications, and to proceed further with the present derivation, it is convenient to rewrite Eq. (2.3) as¹⁴

$$N^{(t)}(E, J) = \langle N_q(E, J, Q) \rangle_{\Omega}, \quad (2.4)$$

$$N_q(E, J, Q) = \frac{1}{2\pi^{4-k_a}} s'^2 \int d^{\nu}\mathcal{P} \delta(H(Q, \mathcal{P}) - E) \times \delta(\hat{\mathbf{J}}(Q, \mathcal{P}) - \mathbf{J}) \dot{s}\Theta(\dot{s}), \quad (2.5)$$

where $\langle \dots \rangle_{\Omega} = \int \dots d\Omega^{(1)} d\Omega^{(2)} d\Omega^{(12)} / \int d\Omega^{(1)} d\Omega^{(2)} d\Omega^{(12)}$ denotes the averaging over all possible orientations of the fragments and the vector $\mathbf{s}$ and $k_a=0,1$ is the number of monatomic fragments. The quantity $N_q(E, J, Q)$ can be loosely viewed as a number of states for a specific configuration Q .

Next we derive an analytical expression for $N_q(E, J, Q)$. To this end it is convenient to rewrite Eq. (2.5) making use of the three-dimensional representation of the one-dimensional delta function in angular momentum,⁸

$$N_q(E, J, Q) = \frac{1}{2\pi^{4-k_a}} s'^2 J^2 \int d^{\nu}\mathcal{P} d\Omega_J \delta(H(Q, \mathcal{P}) - E) \times \delta^3(\hat{\mathbf{J}}(Q, \mathcal{P}) - \mathbf{J}) \dot{s}\Theta(\dot{s}), \quad (2.6)$$

where $\hat{\mathbf{J}}(Q, \mathcal{P})$ is the total angular momentum vector in the laboratory frame and Ω_J denotes the solid angle describing the direction of the vector $\mathbf{J}$ whose length is equal to J .

The Hamiltonian $H(Q, \mathcal{P}) = K(Q, \mathcal{P}) + V(Q)$, where $K(Q, \mathcal{P})$ is the kinetic energy, which is quadratic over the generalized momenta, and $V(Q)$ is the potential energy, which depends only on the coordinates. In the chosen representation, which refers to the molecular frame in which the

inertia tensor is diagonal, the kinetic energy does not depend on the coordinates $\mathcal{Q}$, and takes the following simple form:

$$K(\mathcal{P}) = \frac{\mathbf{p}^2}{2\mu} + \sum_{k=1}^2 \sum_{i=1}^{\nu_k} \frac{J_i^{(k)2}}{2I_i^{(k)}}, \quad (2.7)$$

where $I_i^{(k)}$ denotes the i th principal moment of inertia of the k th fragment. For linear species $I_1 = I_2$.

The total angular momentum vector, $\hat{\mathbf{J}}(\mathcal{Q}, \mathcal{P})$, is linear in the generalized momenta, and is given by

$$\hat{\mathbf{J}}(\mathcal{Q}, \mathcal{P}) = \mathbf{r} \times \mathbf{p} + \sum_{k=1}^2 \sum_{i=1}^{\nu_k} \mathbf{n}_i^{(k)} J_i^{(k)}, \quad (2.8)$$

where $\mathbf{a} \times \mathbf{b}$ denotes a vector cross product and $\mathbf{n}_i^{(k)}$ is the unit vector directed along the i th principal axis of the k th fragment.

The kinetic energy defines a scalar product in the generalized momentum space. It is convenient to renormalize the generalized momenta (and use the same notation $\mathcal{P}$ for new, renormalized values) as follows:

$$\tilde{\mathbf{p}} = \frac{\mathbf{p}}{\sqrt{\mu}}, \quad \tilde{J}_i^{(k)} = \frac{J_i^{(k)}}{\sqrt{I_i^{(k)}}}, \quad k=1,2 \quad i=1,\dots,\nu_k, \quad (2.9)$$

so that K takes the standard form:

$$K(\mathcal{P}) = \sum_{i=1}^{\nu} P_i^2/2. \quad (2.10)$$

This transformation adds a factor of

$$\mu^{3/2} \prod_{k=1}^2 \prod_{i=1}^{\nu_k} \sqrt{I_i^{(k)}} \quad (2.11)$$

to Eq. (2.6). In the new representation the angular momentum vector $\hat{\mathbf{J}}$ is given by

$$\hat{\mathbf{J}}(\mathcal{Q}, \mathcal{P}) = \sqrt{\mu} \mathbf{r} \times \tilde{\mathbf{p}} + \sum_{k=1}^2 \sum_{i=1}^{\nu_k} \sqrt{I_i^{(k)}} \mathbf{n}_i^{(k)} \tilde{J}_i^{(k)}. \quad (2.12)$$

Next, an expression for $\dot{\mathbf{s}} = \dot{\mathbf{s}} \cdot \mathbf{s}/s$, which is also a linear function of the generalized momenta, is obtained. Using Eq. (2.2) the time derivative $\dot{\mathbf{s}}$ can be written as

$$\dot{\mathbf{s}} = \mathbf{v} + \boldsymbol{\omega}^{(1)} \times \mathbf{d}^{(1)} - \boldsymbol{\omega}^{(2)} \times \mathbf{d}^{(2)}, \quad (2.13)$$

where $\boldsymbol{\omega}^{(k)}$ is the rotational frequency of the k th fragment. Taking into account that in the molecular frame $\omega_i^{(k)} = J_i^{(k)}/I_i^{(k)}$ and that $\mathbf{v} = \mathbf{p}/\mu$ one obtains the following expression for $\dot{\mathbf{s}}$:

$$\dot{\mathbf{s}} = \mu^{-1/2} \mathbf{n}^{(12)} \cdot \tilde{\mathbf{p}} + \sum_{k=1}^2 \sum_{i=1}^{\nu_k} (-1)^k (I_i^{(k)})^{-1/2} \times (\mathbf{n}^{(12)}, \mathbf{d}^{(k)}, \mathbf{n}_i^{(k)}) \tilde{J}_i^{(k)}, \quad (2.14)$$

where $(\mathbf{a}, \mathbf{b}, \mathbf{c}) \equiv (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$ and $\mathbf{n}^{(12)}$ is the unit vector in the direction of $\mathbf{s}$.

In order to focus on the linear dependence of both the time derivative of the reaction coordinate, $\dot{\mathbf{s}}$, and the components of the total angular momentum, $\hat{\mathbf{J}}_i$ on the generalized momenta, $\mathcal{P}$, we rewrite Eqs. (2.14) and (2.12) as

$$\dot{\mathbf{s}} = \mathcal{X} \cdot \mathcal{P}, \quad \hat{\mathbf{J}}_i = \mathcal{Y}^{(i)} \cdot \mathcal{P}, \quad i=1,2,3, \quad (2.15)$$

where $\mathcal{A} \cdot \mathcal{B} = \sum_{\alpha=1}^{\nu} A_{\alpha} B_{\alpha}$ denotes a scalar product. The components of the ν -dimensional vectors X_{α} and $Y_{\alpha}^{(i)}$, $\alpha = 1, \dots, \nu$ are easily obtained from Eqs. (2.14) and (2.12), respectively. Then, substituting Eq. (2.15) into Eq. (2.6) [and taking into account the scaling, Eqs. (2.9) and (2.11)], one obtains

$$N_q(E, J, \mathcal{Q}) = F_0 \int d^{\nu} \mathcal{P} d\Omega_J \delta(\mathcal{P} \cdot \mathcal{P}/2 + V(\mathcal{Q}) - E) \times \prod_{i=1}^3 \delta(\mathcal{Y}^{(i)} \cdot \mathcal{P} - J_i) \Theta(\mathcal{X} \cdot \mathcal{P}) \mathcal{X} \cdot \mathcal{P}, \quad (2.16)$$

$$F_0 = \frac{1}{2\pi^{4-k_a}} s'^2 J^2 \mu^{3/2} \prod_{k=1}^2 \prod_{i=1}^{\nu_k} \sqrt{I_i^{(k)}}. \quad (2.17)$$

The δ functions $\delta(\mathcal{Y}^{(i)} \cdot \mathcal{P} - J_i)$ effectively restrict the integration over the generalized momenta $\mathcal{P}$ in Eq. (2.16) to the $(\nu-3)$ -dimensional hyperplane defined by the constraints

$$\mathcal{Y}^{(i)} \cdot \mathcal{P} = J_i, \quad i=1,2,3. \quad (2.18)$$

To integrate them out we first find the minimum of the kinetic energy $K(\mathcal{P}) = \mathcal{P} \cdot \mathcal{P}/2$ in the hyperplane defined by Eq. (2.18). This minimization can be accomplished using the method of undetermined Lagrange multipliers, in which one is looking for an unconstrained minimum of the auxiliary function

$$\tilde{K}(\mathcal{P}) = \mathcal{P} \cdot \mathcal{P}/2 - \sum_{i=1}^3 \omega_i \mathcal{Y}^{(i)} \cdot \mathcal{P}, \quad (2.19)$$

where ω_i are the Lagrange multipliers, which are yet to be defined. It is easy to see that this minimum occurs for $\mathcal{P}^{(J)}$ given by

$$\mathcal{P}^{(J)} = \sum_{i=1}^3 \omega_i \mathcal{Y}^{(i)}. \quad (2.20)$$

The kinetic energy at the minimum $K^{(J)}$ is given by

$$K^{(J)} = \mathcal{P}^{(J)} \cdot \mathcal{P}^{(J)}/2 = \frac{1}{2} \sum_{i,j=1}^3 \omega_i \omega_j I_{i,j}, \quad (2.21)$$

$$I_{i,j} = \mathcal{Y}^{(i)} \cdot \mathcal{Y}^{(j)}. \quad (2.22)$$

The physical meaning of the Lagrange multipliers ω_i , and of the kinetic energy $K^{(J)}$ is elucidated by noting that $\mathcal{Y}^{(i)} \cdot \mathcal{P}$ is the component $\hat{J}_i$, of the total angular momentum [cf. Eq. (2.15)], and that the minimum of the kinetic energy at the constrained total angular momentum $\hat{\mathbf{J}}$ corresponds to that for rotation of the system as a whole. Then, it is clear, that ω_i are the components of the angular frequency for this overall rotation in the laboratory frame. The values of ω_i are obtained from the fact that $\mathcal{P}^{(J)}$ should belong to the hyperplane defined by Eq. (2.18). Substituting Eq. (2.20) into Eq. (2.18) one obtains the following system of equations for ω_i :

$$\sum_{j=1}^3 I_{i,j} \omega_j = J_i, \quad i=1,2,3, \quad (2.23)$$

where $I_{i,j}$ is given by Eq. (2.22). Using Eq. (2.23) the expression for $K^{(J)}$, Eq. (2.21), can be rewritten as

$$K^{(J)}(\mathbf{J}) = \frac{1}{2} \sum_{i,j=1}^3 J_i J_j I_{i,j}^{-1}, \quad (2.24)$$

where I^{-1} denotes the inverse matrix.

The physical meaning of the matrix I can be clarified by substituting Eq. (2.12) into Eq. (2.22), to obtain

$$\begin{aligned} I_{j,l} &= \sum_{k=1}^2 \sum_{i=1}^{\nu_k} I_{i,j}^{(k)} n_{i,j}^{(k)} n_{i,l}^{(k)} + \mu (\delta_{j,l} \mathbf{r}^2 - r_j r_l) \\ &= \sum_{k=1}^2 I_{j,l}^{(k)} + I_{j,l}^{(\text{orb})}, \quad j, l = 1, 2, 3, \end{aligned} \quad (2.25)$$

where $I_{j,l}^{(k)}$ is the j, l th element of the inertia tensor for the k th fragment in the laboratory frame and $I_{j,l}^{(\text{orb})}$ is the j, l th element of the inertia tensor for the orbital motion in the laboratory frame. Thus, one can conclude that I is the inertia tensor for the whole system, and its eigenvalues are the moments of inertia for the whole system.

After shifting the integration variable $\mathcal{P}$ by $\mathcal{P}^{(J)}$ (and using the same notation $\mathcal{P}$ for new, shifted values), Eq. (2.16) takes the form

$$\begin{aligned} N_q(E, J, Q) &= F_0 \int d\nu \mathcal{P} d\Omega_J \Theta(\mathcal{X} \cdot \mathcal{P}) \mathcal{X} \cdot \mathcal{P} \\ &\quad \times \delta(\mathcal{P} \cdot \mathcal{P}/2 + V(Q) + K^{(J)}(\mathbf{J}) - E) \\ &\quad \times \prod_{i=1}^3 \delta(\mathcal{Y}^{(i)} \cdot \mathcal{P}), \end{aligned} \quad (2.26)$$

where we have used the fact that

$$\mathcal{X} \cdot \mathcal{Y}^{(i)} = 0, \quad (2.27)$$

which can be verified by direct calculation, using Eqs. (2.12) and (2.14). Alternatively, the result in Eq. (2.27) can be obtained from the following physical speculation. Any ν -dimensional vector in the momentum space corresponds to a particular type of motion. In particular, the momentum space vector $\mathcal{Y}^{(i)}$ corresponds to the rotation of the system as a whole relative to the i th laboratory frame axis (see the previous discussion). On the other hand, the product $\mathcal{X} \cdot \mathcal{P}$ is the time derivative of the reaction coordinate $\dot{s}$ [cf. Eq. (2.15)] for the motion described by the vector $\mathcal{P}$. Thus, the scalar product $\mathcal{X} \cdot \mathcal{Y}^{(i)}$ can be interpreted as the time derivative of the reaction coordinate for the rotation of the system as a whole relative to the i th laboratory frame axis. But, the reaction coordinate is an internal coordinate, which, of course, does not depend on the orientation of the system or correspondingly on any rotational motion of the system as a whole. As a result, one arrives at Eq. (2.27).

To integrate over the delta functions in the components of the total angular momentum in Eq. (2.26) we make an *orthogonal* transformation of the generalized momenta $\mathcal{P}$, such that the vectors $\mathcal{Y}^{(i)}$ in the new basis belong to its three-dimensional subspace $\mathcal{P}^\perp$. The Jacobian of this transformation is equal to unity due to its orthogonality. In the new basis the vectors $\mathcal{Y}^{(i)}$ can be considered as three-dimensional ones which belong to the $\mathcal{P}^\perp$ subspace. In con-

trast, it is clear from Eq. (2.27) that the vector $\mathcal{X}$ belongs to the $(\nu-3)$ -dimensional subspace $\mathcal{P}^\perp$ orthogonal to $\mathcal{P}^\perp$. Thus, Eq. (2.26) can be rewritten as

$$\begin{aligned} N_q(E, J, Q) &= F_0 \int d^3 \mathcal{P}^\perp \prod_{i=1}^3 \delta(\mathcal{Y}^{(i)} \cdot \mathcal{P}^\perp) \\ &\quad \times \int d^{(\nu-3)} \mathcal{P}^\perp d\Omega_J \Theta(\mathcal{X} \cdot \mathcal{P}^\perp) \mathcal{X} \cdot \mathcal{P}^\perp \\ &\quad \times \delta(\mathcal{P}^\perp \cdot \mathcal{P}^\perp/2 + V(Q) + K^{(J)}(\mathbf{J}) - E). \end{aligned} \quad (2.28)$$

In the new basis the components, $Y_j^{(i)}$, of the vectors $\mathcal{Y}^{(i)}$, and $\mathcal{X}$ are no longer given by Eqs. (2.12) and (2.14). However, the scalar products $I_{i,j}$ of the vectors $\mathcal{Y}^{(i)}$, Eq. (2.22), and the length of the vector $\mathcal{X}$ are preserved because of the orthogonality of the underlying transformation.

To calculate the first integral in Eq. (2.28),

$$F_1 = \int d^3 \mathcal{P}^\perp \prod_{i=1}^3 \delta(\mathcal{Y}^{(i)} \cdot \mathcal{P}), \quad (2.29)$$

(the upper index of $\mathcal{P}^\perp$ is omitted for simplicity of notation) we transform integration variables to $\tilde{\mathcal{P}}$:

$$\tilde{\mathcal{P}}_i = \mathcal{Y}^{(i)} \cdot \mathcal{P} = \sum_{j=1}^3 Y_j^{(i)} \mathcal{P}_j. \quad (2.30)$$

Then F_1 is equal to the determinant of the transformation,

$$F_1 = \left| \frac{\partial(\mathcal{P})}{\partial(\tilde{\mathcal{P}})} \right| = |\det(Y)|^{-1}, \quad (2.31)$$

where we denote by Y the 3 by 3 matrix whose i, j th element is $Y_j^{(i)}$. On the other hand, one can rewrite the expression for the matrix I , Eq. (2.22), in the following form:

$$Y \cdot Y^T = I, \quad (2.32)$$

where Y^T denotes a transposed matrix. Taking the determinant on both sides of Eq. (2.32) one obtains

$$\det(Y)^2 = \det(I).$$

Substituting this equation into Eq. (2.31) one obtains

$$F_1 = \det(I)^{-1/2} = (I_1 I_2 I_3)^{-1/2}, \quad (2.33)$$

where I_i are the eigenvalues of the symmetric operator $I_{i,j}$, Eq. (2.22), i.e., the moments of inertia for the system as a whole.

The second integral F_2 in Eq. (2.28) can be written as

$$F_2 = \int d\Omega_J d^{(\nu-3)} \mathcal{P}^\perp \Theta(\mathcal{X} \cdot \mathcal{P}) \mathcal{X} \cdot \mathcal{P} \delta(\mathcal{P} \cdot \mathcal{P}/2 - E^{(\text{int})}), \quad (2.34)$$

$$E^{(\text{int})} = E - V(Q) - K^{(J)}(\mathbf{J}) \quad (2.35)$$

(the upper index $\perp$ of $\mathcal{P}^\perp$ is again omitted for simplicity of notation), where $E^{(\text{int})}$ denotes the kinetic energy of the “internal degrees of freedom.” Taking into account that the integration over $d\Omega_J$ in Eq. (2.34) is invariant under an arbitrary rotation of the system of coordinates and using the definition of $K^{(J)}(\mathbf{J})$, Eq. (2.24), one can rewrite Eq. (2.35) without loss of generality as

$$E^{(\text{int})} = E - V(\mathcal{Q}) - \sum_{i=1}^3 J_i^2/2I_i, \quad (2.36)$$

where I_i are again the moments of inertia of the whole system.

To integrate over $\mathcal{P}$ in Eq. (2.34) another orthogonal transformation is made, such that the first basis vector will be collinear with $\mathcal{X}$. Then the corresponding variable can be integrated out yielding

$$F_2 = \sqrt{\mathcal{X} \cdot \mathcal{X}} \int d\Omega_J d^{(\nu-4)}\mathcal{P} \Theta(E^{(\text{int})} - \mathcal{P} \cdot \mathcal{P}/2), \quad (2.37)$$

where $\mathcal{P}$ now denotes the remaining unintegrated variables. Noting that the integral over $\mathcal{P}$ corresponds to the volume of a $(\nu-4)$ -dimensional sphere with radius $\sqrt{2E^{(\text{int})}}$, one obtains

$$F_2 = \frac{(2\pi)^{\nu/2-2}}{\Gamma(\nu/2-1)} \sqrt{\mathcal{X} \cdot \mathcal{X}} \int d\Omega_J \max(E^{(\text{int})}, 0)^{\nu/2-2}. \quad (2.38)$$

In Eq. (2.38) the factor $\Gamma(x)$ is the gamma function, which arises from the expression for the volume, V_n , of the n -dimensional sphere of radius r , $V_n(r) = r^n \pi^{n/2}/\Gamma(n/2 + 1)$.

Collecting Eqs. (2.17), (2.33), and (2.38) back into Eq. (2.28) and substituting Eq. (2.14) into it, one obtains the following expression for $N_q(E, J, \mathcal{Q})$:

$$N_q(E, J, \mathcal{Q}) = \Phi J^2 \mu s'^2 \frac{4\pi^{k_a-2}}{\Gamma(\nu/2-1)} \frac{\prod_{k=1}^2 \prod_{i=1}^{\nu_k} \sqrt{2\pi I_i^{(k)}}}{\prod_{i=1}^3 \sqrt{2\pi I_i(\mathcal{Q})}} \times \left\langle \max \left(E - V(\mathcal{Q}) - \sum_{i=1}^3 J_i^2/2I_i(\mathcal{Q}), 0 \right)^{\nu/2-2} \right\rangle_{\Omega_J}, \quad (2.39)$$

$$\Phi = \sqrt{\mu \mathcal{X} \cdot \mathcal{X}} = \sqrt{1 + \mu \sum_{k=1}^2 \sum_{i=1}^{\nu_k} (\mathbf{n}^{(12)}, \mathbf{d}^{(k)}, \mathbf{n}_i^{(k)})^2 / I_i^{(k)}}, \quad (2.40)$$

where $(\mathbf{a}, \mathbf{b}, \mathbf{c}) \equiv \mathbf{a} \times \mathbf{b} \cdot \mathbf{c}$ and $\langle \cdots \rangle_{\Omega_J}$ denotes the averaging over all possible orientations of the vector $\mathbf{J}$, while its length is fixed. In Eq. (2.39) I_i , $i=1,2,3$ denote the moments of inertia for the system as a whole, $I_i^{(k)}$, $k=1,2$, $i=1, \dots, \nu_k$ are the moments of inertia of the fragments, μ is the reduced mass, $\mathbf{n}^{(12)}$ is the unit vector directed from the second pivot point to the first one, $\mathbf{d}^{(k)}$, $k=1,2$ is the vector connecting the center of mass of the k th fragment to its pivot point, $\mathbf{n}_i^{(k)}$ is the unit vector directed along the i th principal axis of the k th fragment, and $k_a=0,1$ is the number of monatomic fragments. It is also stressed in Eq. (2.39) that the total moments of inertia I_i , $i=1,2,3$ are not constants but are configuration dependent.

Equation (2.39) together with Eq. (2.4) are the main result of this paper. They provide a straightforward means for evaluating the transitional mode contribution to the energy and total angular momentum resolved reactive flux, Eq. (2.1), through the dividing surface defined by a fixed distance between two pivot points. Equation (2.39) differs from a similar result of Smith¹⁵ in the form of the kinematic factor

Φ , Eq. (2.40), with the present form being more convenient for numerical evaluation. In Sec. III we consider an application of these equations to the reaction of the vinyl radical with the hydrogen atom.

The kinematic factor Φ , Eq. (2.40), contains contributions both from the relative translational motion of the fragments and from rotational motion of either fragment. The contribution from the fragment rotational motion has a quadratic dependence on the displacement of each pivot point from its center of mass. Furthermore, this contribution, and the overall factor are clearly minimized when the pivot point of each fragment coincides with its center of mass.

While the kinematic factor Φ , Eq. (2.40), is valid for both linear and nonlinear molecules it is convenient to rewrite the term for rotational motion of the linear fragment in a different form:

$$\mu I^{-1} \sum_{i=1}^2 (\mathbf{n}^{(12)}, \mathbf{d}, \mathbf{n}_i)^2 = \mu I^{-1} d^2 [1 - (\mathbf{n}^{(12)} \cdot \mathbf{n}_0)^2], \quad (2.41)$$

where $\mathbf{n}_0$ is the unit vector directed along the axis of the molecule, $I=I_1=I_2$ is its moment of inertia, and we have used the fact that $\mathbf{d}$ is collinear with $\mathbf{n}_0$.

For completeness, we now derive analogous results for the transitional mode contributions to the reactive flux in the canonical and microcanonical cases. For the canonical case, the transitional mode contribution to the reactive flux is given by

$$Z(T) = (2\pi)^{-\nu} \int d^\nu \mathcal{Q} d^\nu \mathcal{P} e^{-H(\mathcal{Q}, \mathcal{P})/T} \delta(s(\mathcal{Q}) - s') \dot{s} \Theta(\dot{s}), \quad (2.42)$$

where T is the temperature in the energy units. Since the delta function over the total angular momentum does not enter into the definition of Z , the derivation of the expression for it is much simpler. However, one can circumvent even that simplified derivation by noticing that $Z(T)$ can be expressed as

$$Z(T) = \int dE N(E) e^{-E/T}, \quad N(E) = \int_0^\infty dJ N(E, J), \quad (2.43)$$

where $N(E)$ is the microcanonical number of states,

$$N(E) = (2\pi)^{-\nu} \int d^\nu \mathcal{Q} d^\nu \mathcal{P} \delta(H(\mathcal{Q}, \mathcal{P}) - E) \times \delta(s(\mathcal{Q}) - s') \dot{s} \Theta(\dot{s}). \quad (2.44)$$

Integrating Eq. (2.39) first over J and substituting the result into Eq. (2.4) one obtains the following expression for $N(E)$:

$$N(E) = \mu s'^2 \frac{\pi^{k_a-3}}{\Gamma((\nu+1)/2)} \prod_{k=1}^2 \prod_{i=1}^{\nu_k} \sqrt{2\pi I_i^{(k)}} \times \langle \Phi \max(E - V(\mathcal{Q}), 0)^{(\nu-1)/2} \rangle_{\Omega}, \quad (2.45)$$

where we have switched between polar and Cartesian coordinates while integrating over $d\Omega_J J^2 dJ$ and have used the identity $\int_{-1}^1 (1-x^2)^{n/2-1} dx = \sqrt{\pi} \Gamma(n/2) / \Gamma((n+1)/2)$.

Integrating Eq. (2.45) over E with the factor of $e^{-E/T}$ yields the following expression for Z :

$$Z = T^{(\nu+1)/2} \mu s'^2 \pi^{k_a-3} \prod_{k=1}^2 \prod_{i=1}^{\nu_k} \sqrt{2\pi I_i^{(k)}} \langle \Phi e^{-V(Q)/T} \rangle_{\Omega}. \quad (2.46)$$

B. Elliptical dividing surface

The expression for the E, J -resolved reactive flux, Eq. (2.39), was derived under the assumption that the dividing surface is defined by a fixed distance between the two pivot points. This derivation can easily be applied, with minor modifications, to more general surfaces via the simple consideration of the changes in the definitions of s and $\dot{s}$. We illustrate this point for the elliptical dividing surface when one of the fragments is a nonlinear molecule and the other is an atom. The dividing surface is defined as

$$l = \mathbf{r} - \mathbf{d}, \quad l = l \mathbf{n}^{(12)}, \quad (2.47)$$

$$\sum_{\alpha=1}^3 (\mathbf{l} \cdot \mathbf{n}_{\alpha}^{(e)})^2 A_{\alpha} = l^2 \sum_{\alpha=1}^3 (\mathbf{n}^{(12)} \cdot \mathbf{n}_{\alpha}^{(e)})^2 A_{\alpha} = s^2, \quad (2.48)$$

where $\mathbf{l}$ is the vector connecting the center of the ellipsoid to the atomic fragment, $\mathbf{r}$ is the vector connecting the centers of masses of the two fragments, $\mathbf{d}$ is the vector connecting the center of mass of the molecular fragment and the center of the ellipsoid, which is fixed in the molecular frame of the nonatomic fragment, $\mathbf{n}_{\alpha}^{(e)} \alpha=1,2,3$ are the directions of the axes of the ellipsoid, which are fixed in the molecular frame of the nonatomic fragment, A_{α} are the stretching factors along the corresponding axes of the ellipsoid, and s is the reaction coordinate which defines the ellipsoid size. It is convenient to use s and $\mathbf{n}^{(12)}$ as independent orbital coordinates. The Jacobian of the transformation from $\mathbf{r}$ to $\mathbf{l}$ is unity and the Jacobian of the transformation from Cartesian coordinates $\mathbf{l}$ to angular coordinates $\mathbf{n}^{(12)}$ and s is $l^2 \partial l / \partial s$ (the partial derivative is taken at fixed angular coordinates), which provides an additional factor of $l^2 / s^2 \partial l / \partial s$ in Eq. (2.5) for the considered dividing surface. It follows from Eq. (2.48) that $\partial l / \partial s$ is equal to

$$\frac{\partial l}{\partial s} = \frac{l}{s} = \frac{1}{\sqrt{\sum_{\alpha=1}^3 A_{\alpha} (\mathbf{n}^{(12)} \cdot \mathbf{n}_{\alpha}^{(e)})^2}}. \quad (2.49)$$

Taking the time derivative of Eq. (2.48) one obtains the following expression for $\dot{s}$ in terms of the generalized momenta, Eq. (2.9),

$$\begin{aligned} \dot{s} = & \frac{1}{\sqrt{\sum_{\alpha=1}^3 A_{\alpha} (\mathbf{n}^{(12)} \cdot \mathbf{n}_{\alpha}^{(e)})^2}} \left\{ \mu^{-1/2} \sum_{\alpha=1}^3 A_{\alpha} (\mathbf{n}^{(12)} \cdot \mathbf{n}_{\alpha}^{(e)}) \right. \\ & \times (\mathbf{n}_{\alpha}^{(e)} \cdot \tilde{\mathbf{p}}) + \sum_{i=1}^3 \left[\sum_{\alpha=1}^3 A_{\alpha} (\mathbf{n}^{(12)} \cdot \mathbf{n}_{\alpha}^{(e)}) \right. \\ & \left. \left. \times (\mathbf{n}_i^{(1)} \cdot \mathbf{n}_{\alpha}^{(e)}, \mathbf{r}) \right] \tilde{J}_i^{(1)} / \sqrt{I_i^{(1)}} \right\}, \end{aligned} \quad (2.50)$$

where $I_i^{(1)}$, $\mathbf{n}_i^{(1)}$ are the moments of inertia and the principal axes of the molecular fragment, respectively. Equation (2.50) is a substitute for Eq. (2.14) for this elliptical dividing surface. It provides new expressions for the components of the vector $\mathcal{X}$, Eq. (2.15). Otherwise the derivation is unchanged. In particular, one still has $\mathcal{X} \cdot \mathcal{Y}^{(i)} = 0$ because s is again an internal coordinate. As a result, the expression for the E, J -resolved reactive flux is still given by Eq. (2.39), but with the kinematic factor Φ , which was given by Eq. (2.40), replaced by

$$\begin{aligned} \Phi = & \frac{l^2}{s^2} \frac{\partial l}{\partial s} \sqrt{\mu \mathcal{X} \cdot \mathcal{X}} \\ = & \left[\sum_{\alpha=1}^3 A_{\alpha} (\mathbf{n}^{(12)} \cdot \mathbf{n}_{\alpha}^{(e)})^2 \right]^{-2} \left\{ \sum_{i=1}^3 \left[\sum_{\alpha=1}^3 A_{\alpha} (\mathbf{n}^{(12)} \cdot \mathbf{n}_{\alpha}^{(e)}) n_{\alpha,i}^{(e)} \right]^2 \right. \\ & \left. + \sum_{i=1}^3 \left[\sum_{\alpha=1}^3 A_{\alpha} (\mathbf{n}^{(12)} \cdot \mathbf{n}_{\alpha}^{(e)}) (\mathbf{n}_i^{(1)} \cdot \mathbf{n}_{\alpha}^{(e)}, \mathbf{r}) \right]^2 \mu / I_i^{(1)} \right\}^{1/2}. \end{aligned} \quad (2.51)$$

C. Planar dividing surface

When the reactive system consists of a nonlinear fragment and an atom it is convenient sometimes to use a planar surface for a portion of the dividing surface. While, strictly speaking, Eqs. (2.4) and (2.39) are not valid for a planar surface, they can be easily adapted to this case. To this end, we first define the orientation of the planar surface as being perpendicular to a unit vector $\mathbf{n}^{(0)}$ fixed in the molecular frame of the nonlinear fragment. Then, the reaction coordinate s can be defined in terms of the displacement of the plane in the direction of $\mathbf{n}^{(0)}$,

$$s = \mathbf{n}^{(0)} \cdot \mathbf{r}, \quad (2.52)$$

where, as before, $\mathbf{r}$ is the vector connecting the center of mass of the nonlinear fragment and the atomic fragment. Now, instead of using angular coordinates for the relative translational motion as integration variables, the s coordinate and two in-plane coordinates x and y can be used. The averaging over angular orientations of the nonlinear fragment can be excluded due to the isotropy of space. As a result, Eq. (2.4) is substituted by

$$N^{(t)}(E, J) = \frac{1}{4\pi} \int dx dy N_q(E, J, Q), \quad (2.53)$$

where the expression for $N_q(E, J, Q)$ is given by an equation similar to Eq. (2.1) but without the s'^2 factor and the laboratory frame is assumed to coincide with the molecular frame of the nonlinear fragment.

Further derivation proceeds in the same way as before. Taking the time derivative of Eq. (2.52) the expression for $\dot{s}$ is obtained

$$\dot{s} = \mathbf{n}^{(0)} \cdot \mathbf{v} + \boldsymbol{\omega}^{(1)} \times \mathbf{n}^{(0)} \cdot \mathbf{r}, \quad (2.54)$$

which can be rewritten in terms of the renormalized momenta, Eq. (2.9), as

$$\dot{s} = \mu^{-1/2} \mathbf{n}^{(0)} \cdot \tilde{\mathbf{p}} + \sum_{i=1}^3 (I_i^{(1)})^{-1/2} (\mathbf{n}^{(0)}, \mathbf{r}, \mathbf{n}_i^{(1)}) \tilde{J}_i^{(1)}. \quad (2.55)$$

Using Eq. (2.55) one arrives at an expression for $N_q(E, J, Q)$ similar to Eq. (2.39) but in which s'^2 is omitted

$$N_q(E, J, Q) = \Phi J^2 \mu \frac{4}{\pi} \prod_{i=1}^3 \sqrt{I_i^{(1)}/I_i(Q)} \times \left\langle \max \left(E - V(Q) - \sum_{i=1}^3 J_i^2/2I_i(Q), 0 \right) \right\rangle_{\Omega_J}, \quad (2.56)$$

and with the kinematic factor Φ now given by

$$\Phi = \sqrt{1 + \mu \sum_{i=1}^3 (\mathbf{n}^{(0)}, \mathbf{r}, \mathbf{n}_i^{(1)})^2 / I_i^{(1)}}. \quad (2.57)$$

III. APPLICATION

A. Methodology

In this section we consider the results of a numerical application of the expressions for the reactive flux, Eqs. (2.4), (2.39), (2.45), and (2.46), to the addition reaction of a vinyl radical with a hydrogen atom to form ethylene,



These comparisons each use the transitional mode analytic potential from Ref. 12, which was based on a fit to multireference configuration interaction calculations employing a

correlation-consistent polarized valence double zeta basis set. For simplicity, no basis set corrections are included, although they were an important aspect of the study in Ref. 12, and only the ground singlet electronic state is considered. Furthermore, we restrict our attention to solely the transitional modes, since the conserved mode contributions to the transition state and reactant partition functions can be shown to be equivalent under reasonable limiting assumptions.^{16,17}

For completeness, we begin with a brief summary of the methods employed in the required orientational averaging over the solid angles $\Omega^{(i)}$, $i=1,2$ and $\Omega^{(12)}$ in Eqs. (2.4), (2.45), and (2.46). As in our prior work,⁸ the crude Monte Carlo sampling method is used. Here, however, instead of the Euler angle representation for the orientation of the non-linear fragments, we find it more convenient, and likely more efficient, to use quaternion coordinates χ , η , ξ , and ζ .¹⁸ These coordinates can be expressed in terms of Euler angles ϑ , φ , and ψ as¹⁹

$$\begin{aligned} \chi &= \cos(\vartheta/2) \cos(\varphi + \psi)/2, \\ \eta &= \sin(\vartheta/2) \cos(\varphi - \psi)/2, \\ \xi &= \sin(\vartheta/2) \sin(\varphi - \psi)/2, \\ \zeta &= \cos(\vartheta/2) \sin(\varphi + \psi)/2. \end{aligned} \quad (3.2)$$

However, this relation is never used. Instead, the absolute spatial positions of each of the atoms (which is required for the evaluation of the transitional mode potential and moments of inertia, etc.) are obtained through the appropriate application of rotation matrices. Importantly, the rotation matrix A can be directly obtained from the quaternions via¹⁹

$$A = \begin{pmatrix} \chi^2 + \eta^2 - \xi^2 - \zeta^2 & 2(\eta\xi + \chi\zeta) & 2(\eta\zeta - \chi\xi) \\ 2(\eta\xi - \chi\zeta) & \chi^2 - \eta^2 + \xi^2 - \zeta^2 & 2(\xi\zeta + \chi\eta) \\ 2(\eta\zeta + \chi\xi) & 2(\xi\zeta - \chi\eta) & \chi^2 - \eta^2 - \xi^2 + \zeta^2 \end{pmatrix}. \quad (3.3)$$

The four quaternion coordinates are not independent but are restricted to the four-dimensional sphere with unit radius, $\chi^2 + \eta^2 + \xi^2 + \zeta^2 = 1$. Intuitively, it seems plausible to suggest that the averaging over random orientations of the molecule may be substituted by the averaging over the surface of this sphere. In the Appendix it is shown that it is indeed so. Thus, the random sampling over the orientation of the non-linear molecule is accomplished via a random sampling over the direction of the four-dimensional vector. Related ideas, but for a three-dimensional vector, are employed in the orientational sampling over the vector $\mathbf{s}$ and for the orientation of linear components such as the line-of-centers and/or linear fragments.

The averaging over the orientation of the $\mathbf{J}$ vector in Eq. (2.39) requires the integration over the surface of the unit sphere, which could also be accomplished via random sampling. It is more efficient, however, to use a combination of numerical and analytic integrations. We use the polar angle

representation and integrate first analytically over the polar angle θ and then numerically over the azimuthal angle ϕ in a similar fashion to that described in Ref. 20.

For the present E , J -resolved variable reaction coordinate (VRC)-TST analysis the pivot point for the H atom is taken to coincide with the H atom itself. That for the vinyl radical is varied over wide ranges, but always taken to lie in its molecular plane. The minimization of the net flux with respect to the location of the pivot point for the vinyl radical as well as to the distance between it and the hydrogen atom provides the optimal E , J -resolved flux, which is then numerically integrated over E and J . In contrast, the canonically optimized reactive flux is obtained by minimizing the reactive flux *after* integration over E and J , which is done analytically, Eq. (2.46). Meanwhile, the microcanonical optimized reactive flux is an intermediate case, where the minimization involves the J integrated flux of Eq. (2.44). In each instance, the rate constants are obtained via normalization by

the partition functions for the reactants, and multiplication by the electronic degeneracy factor $g_e = 1/4$ and the symmetry correction factor $\sigma/\sigma^\ddagger = 1$.

For comparison purposes we have also performed trajectory simulations of the capture process employing Keck's method of initiating trajectories at an approximate transition state dividing surface.³ A brief description of our implementation of this methodology has been provided in Ref. 5 and a more detailed exposition will be provided in a forthcoming article.²¹ For now we simply note that the trajectories were initiated at the canonically optimized dividing surface and were deemed to have reacted upon reaching a certain minimal distance (2 bohr for the frontside channel and 1.5 bohr for the backside channel, cf. Sec. III B for a description of these two channels) from the local potential minimum corresponding to the considered channel, and to have returned to reactants upon reaching center-of-mass separations of 13 bohr or greater.

For the frontside channel the trajectory results are at most weakly dependent on modest variations of the choice for the minimal separation. However, for the backside channel, due to the relatively weak attraction along its minimum energy path, there is a significant dependence on the minimal separation, particularly at higher temperatures. At 1500 K, using a minimal separation of 1.3 bohr decreases the estimate by about 4%. Thus, the results for the backside channel should be viewed with some caution.

B. Dividing surfaces

The transitional mode potential from Ref. 12 for the interaction of the H atom with the vinyl radical is illustrated in Fig. 2. From Fig. 2 it can be seen that there are two barrierless routes for the association reaction, one from each side of the CH portion of the CH_2CH radical. The approach from the side that does not have the H atom is more attractive, and has been termed frontside addition, while approach from the other, less attractive, side has been termed backside addition.¹² The third shallow potential minimum seen in Fig. 2 corresponds to the direct hydrogen abstraction, $\text{C}_2\text{H}_3 + \text{H} \rightarrow \text{C}_2\text{H}_2 + \text{H}_2$, and is of minor importance to the present analysis.

Within the VRC-TST framework a generic dividing surface for the present system is characterized by three parameters: the dividing surface radius s and the in-plane y and z coordinates of the vinyl radical pivot point. The out-of-plane coordinate x of the pivot point is set to 0 due to the symmetry of the system. Different regions in this three-dimensional parameter space correspond to different processes. Thus, surfaces which include only the frontside potential minimum correspond to the frontside addition reaction, while surfaces which include only the backside potential minimum correspond to the backside addition. Meanwhile, surfaces which include both minima correspond to the overall reaction. To estimate a variationally optimized reaction rate constant for the particular process one needs to minimize the reactive flux in the corresponding region of the three-dimensional parameter space of s , y , and z .

Practically, the flux minimization is performed on a finite set of dividing surfaces and the right choice of surfaces

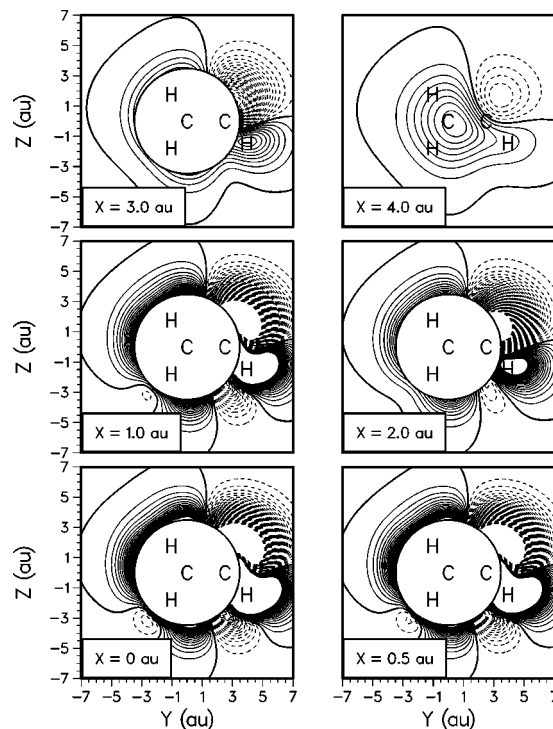


FIG. 2. Contour plots of the potential energy surface for the interaction of C_2H_3 with H. For each plot the H atom is restricted to a plane parallel to the symmetry plane of the vinyl radical, with the coordinate X denoting the distance of the plane from the symmetry plane. Solid contours are positive, dashed contours are negative, and the zero energy contour is shown with a heavy solid line. The contour increment is 1.0 kcal/mol and all distances are in bohr.

in the set can considerably reduce the computational work. The guiding principle is that the optimal dividing surface should avoid as much as possible attractive regions of the potential while maintaining a small surface area. Typically, the reaction coordinate s is varied over a wide range of values (e.g., 4–13 bohr) in a more or less system independent manner. In contrast, the optimal pivot points are very system dependent, but do generally range from about the center of mass to a distance of about 3 bohr from the atom involved in bond formation. Here, such considerations are useful and appropriate in considering the optimal dividing surfaces for the total flux. However, for the present considerations of separate frontside and backside channels, the need to avoid the opposite channel leads to significant deviations from such general observations. In essence, the required avoidance of the opposite minima introduces artificial deviations of these spherically constrained dividing surfaces from more standard results. In reality, the optimal channel specific dividing surface is far from spherical.

Three sets of dividing surfaces have been considered. In the first set the overall flux was optimized without discrimination between different channels. To this end only those surfaces were considered which enclose both the potential minima corresponding to the frontside and backside reaction channels. From the shape of the potential which is shown in Fig. 2 it is clear that the optimal pivot points should be somewhere in between the potential minima of the frontside and the backside reactive channels, closer to the frontside

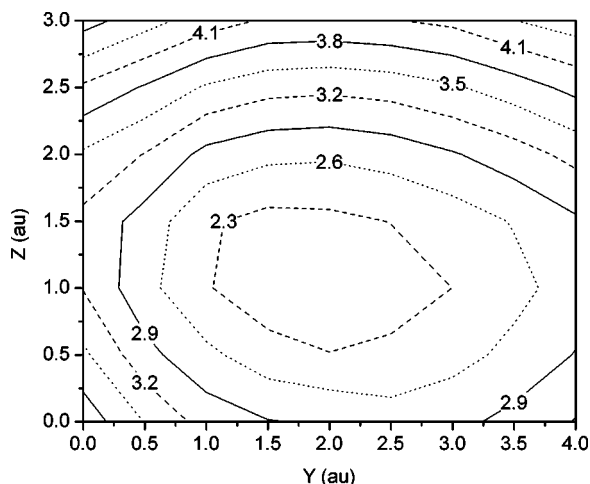


FIG. 3. Contour plot of the bothsides canonical TST rate constant (in units of $10^{-10} \text{ cm}^3/\text{s}$) as a function of the position of the pivot point at 400 K. The rate constant is minimized with respect to the dividing surface radius s .

one. Specifically, the vinyl pivot points in this set were positioned on the grid $0 \leq y \leq 4$, $\Delta y = 0.5$, and $0 \leq z \leq 3$, $\Delta z = 0.5$.²² The dividing surface radius s was varied in the range $4 \leq s \leq 13$ with a step of 0.25 and with the condition that both the frontside and backside potential minima are inside the surface.

It was found that the position of the pivot point for the transition state dividing surface in this case is close to $y = 2$, $z = 1$ for most temperatures, energies, and angular momenta. Such dividing surfaces avoid the contribution to the flux from the direct hydrogen abstraction channel and so this channel will not be considered further here. These calculations are illustrated in Fig. 3, in which the canonical TST rate constant optimized over the dividing surface radius s is shown as a function of the position of the pivot point for 400 K.²³

Although, for this reaction, there is no difference in the products arising from the two separate reaction paths, that is not always case. When the products are different it is often of considerable interest to estimate the branching between them. Furthermore, the optimization of separate transition state dividing surfaces for each individual channel may provide lower and thus better estimates for the overall reactive flux. Thus, we consider two other calculations in which we discriminate the contribution to the rate from different reaction channels. In the second set only those dividing surfaces are considered which enclose the potential minimum corresponding to the frontside channel and do not enclose the one for the backside channel. For the third set the situation is reversed.

The pivot points in the frontside channel set were positioned on the grid $0.5 \leq y \leq 6$, $\Delta y = 0.5$, $2 \leq z \leq 6$, $\Delta z = 0.5$. The radius s was varied in the range $z \leq s \leq z + 4$ with a step of 0.1 in order to separate the frontside and backside reaction channels. These calculations are illustrated in Fig. 4, in which the canonical TST rate constant optimized over the dividing surface radius s is shown as a function of the position of the pivot point for 400 K.²³

The pivot points for the backside channel were posi-

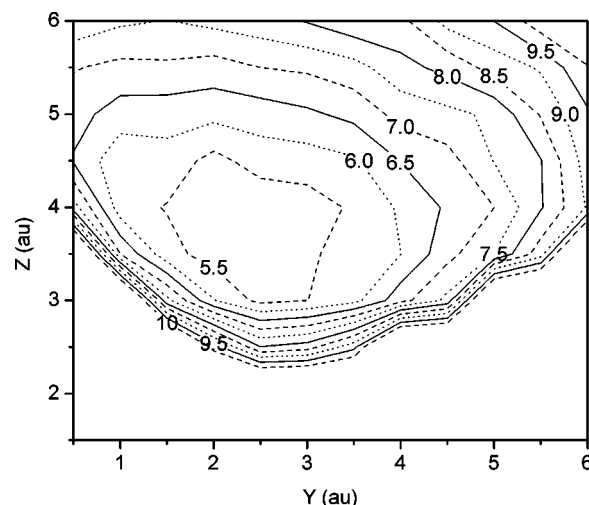


FIG. 4. Contour plot of the frontside canonical TST rate constant (in units of $10^{-10} \text{ cm}^3/\text{s}$) as a function of the position of the pivot point at 400 K. The rate constant is minimized with respect to the dividing surface radius s .

tioned on the grid $0 \leq y \leq 4$, $\Delta y = 0.5$, $-6 \leq z \leq -2$, $\Delta z = 0.5$ and the radius was varied in the range $-z - 2 \leq s \leq -z + 1$ with the step of 0.1. Again, the main idea was to separate the backside and the frontside channel as much as possible. These calculations are illustrated in Fig. 5, in which the canonical TST rate constant optimized over the dividing surface radius s is shown as a function of the position of the pivot point for 400 K.²³

C. Results and discussion

In Fig. 6 the results of the VRC-TST and trajectory calculations of the overall rate constant for the $\text{C}_2\text{H}_3 + \text{H} \rightarrow \text{C}_2\text{H}_4$ reaction are shown. It can be seen that the E, J -resolved optimization provides about 20% improvement over the canonical one. Furthermore, and more importantly, the E, J -resolved VRC-TST results are only about 10%–20%

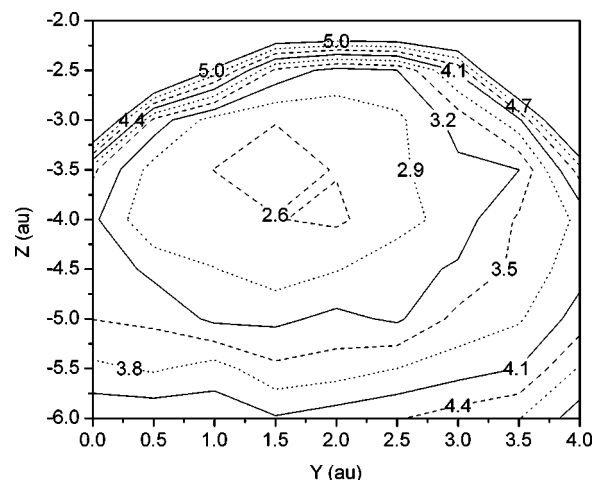


FIG. 5. Contour plot of the backside canonical TST rate constant (in units of $10^{-10} \text{ cm}^3/\text{s}$) as a function of the position of the pivot point at 400 K. The rate constant is minimized with respect to the dividing surface radius s .

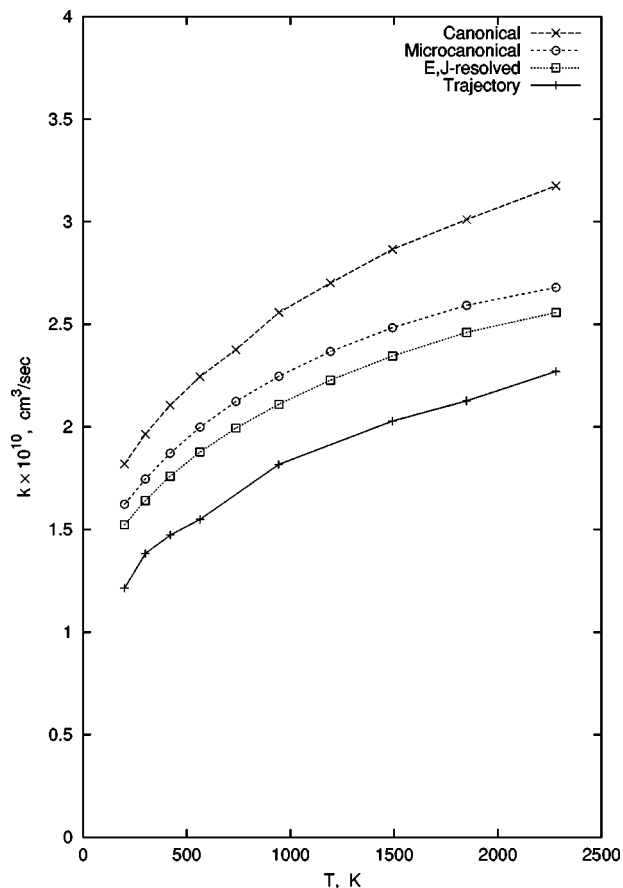


FIG. 6. The overall rate constant for the reaction $\text{C}_2\text{H}_3 + \text{H} \rightarrow \text{C}_2\text{H}_4$ is shown as a function of temperature. The results are presented for canonical, microcanonical, and E, J -resolved VRC-TST optimizations, as well as for trajectory simulations.

greater than the trajectory results. A similar level of agreement between VRC-TST and trajectory simulations was recently found for the $\text{CH}_3 + \text{H}$ reaction.⁵

The situation is dramatically different for the individual frontside and backside rates, Figs. 7 and 8, respectively. At low temperatures, $T \lesssim 500$ K, the canonical optimized frontside and backside rates differ from their microcanonical and E, J -resolved analogs by orders of magnitude, which renders the canonical estimates for these rates at low temperature worthless. The explanation for this failure is in the fact that the spherical dividing surface defined by the distance between the pivot points of the two fragments is not flexible enough to cleanly separate the attractive regions of the interfragment interaction potential corresponding to the different reaction channels. As a result, a small (in volume) region of the dividing surface with large negative interaction energy, makes a dominant contribution to the canonical reactive flux, due to the exponential dependence on the interaction energy, Eq. (2.46). Interestingly, this failure for the individual channel canonical rates for the vinyl + hydrogen reaction contrasts sharply with the results of Robertson *et al.*²⁴ for the $\text{CFH}_2 + \text{H}$ reaction, where it was found that the sum of the partial rates through the different reaction channels improved the estimate of the overall canonical reaction rate for that reaction on a model potential energy surface.

The effect of a strong, but local attractive potential is

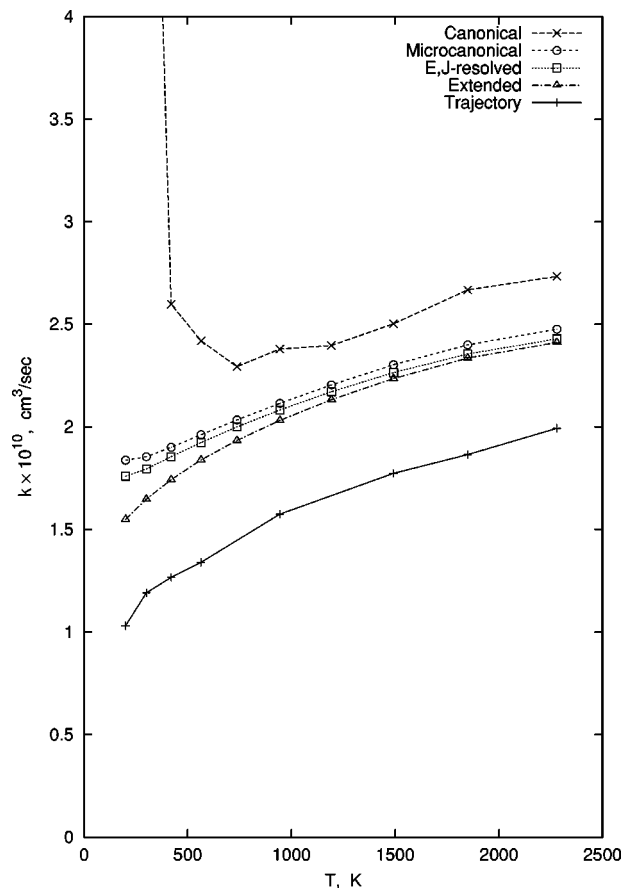


FIG. 7. The frontside rate constant for the reaction $\text{C}_2\text{H}_3 + \text{H} \rightarrow \text{C}_2\text{H}_4$ is shown as a function of temperature. The results are presented for canonical, microcanonical, and E, J -resolved VRC-TST optimizations, for E, J -resolved VRC-TST optimizations with the extended set of dividing surfaces (see the text), and for trajectory simulations.

much less important for microcanonical and E, J -resolved optimizations, since in these cases the energy comes into the expressions for the reactive flux in the $5/2$ and first power, correspondingly [cf. Eqs. (2.39) and (2.45)]. Furthermore, the integration over energy in the microcanonical and E, J -resolved cases only extends down to $E=0$ (i.e., the energy of the separated fragments). As a result, the contribution from the small attractive region on the dividing surface to the microcanonical and E, J -resolved reactive fluxes provides only a minor modification to the total flux. In contrast, the canonical integral implicitly includes all energies down to the minimum potential energy.

The E, J -resolved optimized estimate for the frontside rate constant, while generally being close to or lower than the corresponding overall rate from Fig. 6, is actually slightly larger at the lowest temperatures. The reason for this is related to the expansion of the dividing surface with increasing angular momentum J at a given value of E . As a result, for small values of E and sufficiently large values of J the reactive flux through the dividing surface surrounding the whole molecule happens to be less than the flux through the best dividing surface for just the frontside reaction channel. This effect is even more pronounced for the backside reaction channel, which appears in a bigger deviation of the E, J -resolved rate constant from the result of trajectory simu-

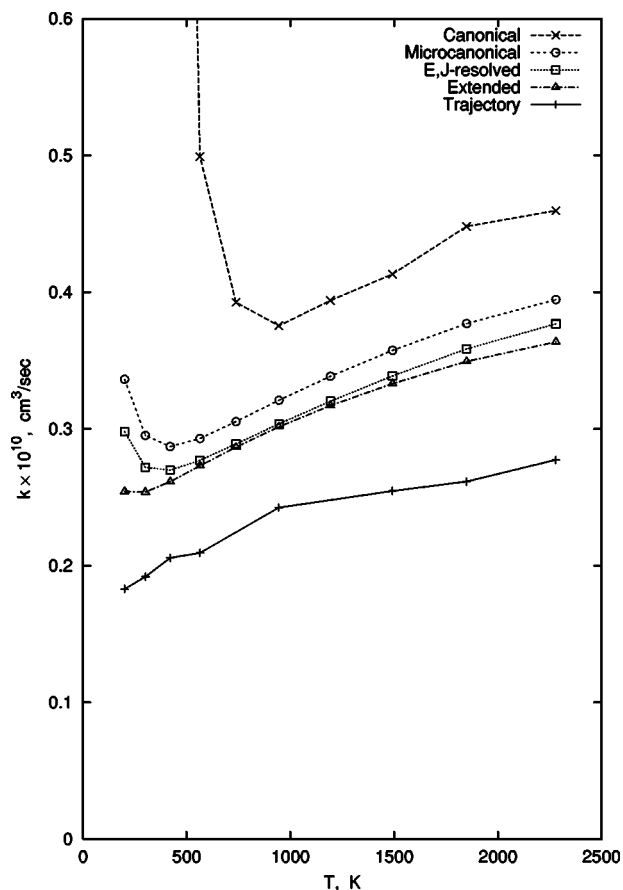


FIG. 8. The backside rate constant for the reaction $C_2H_3 + H \rightarrow C_2H_4$ is shown as a function of temperature. The results are presented for canonical, microcanonical, and E, J -resolved VRC-TST optimizations, for E, J -resolved VRC-TST optimizations with the extended set of dividing surfaces (see the text), and for trajectory simulations.

lations, cf. Fig. 8. To improve the estimates for both the frontside and backside E, J -resolved rate constants extended sets of dividing surfaces are considered. The frontside (backside) extended set includes surfaces from the above-discussed frontside (backside) set as well as the ones from the bothsides set. The corresponding frontside and backside reaction constants are shown as functions of temperature in Figs. 7 and 8, respectively. This extension gives an additional improvement of 10% at low temperatures.

The difference between these extended channel specific E, J -resolved VRC-TST and trajectory rate constants is still somewhat greater than it was for the overall rate constant. Indeed, for the frontside channel the VRC-TST estimate is as much as 50% greater than the trajectory one at low temperatures. Notably, the sum of the frontside and backside rate constants always exceeds that from the dividing surfaces which surround the whole molecule, again in contrast with the results of Robertson *et al.*²⁴ for the $CF_2H + H$ reaction. These observations are indicative of the inadequacy of the single spherical dividing surfaces for separating the total flux into channel specific values. Such surfaces double count the reactive flux in the regions of the overlap of their inner volumes.

An alternative approach is to assume an infinite potential barrier separating the two reactive channels.¹² For the

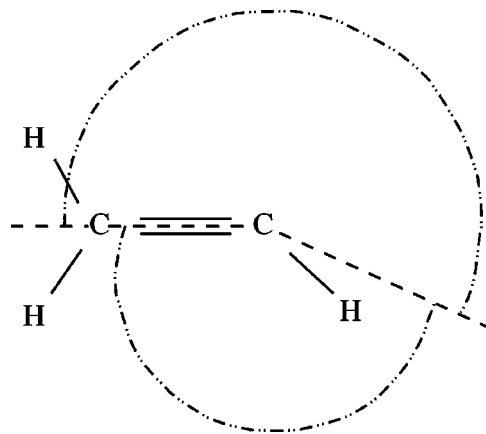


FIG. 9. Cartoon showing the dividing surfaces for the frontside and backside channels for the $C_2H_3 + H \rightarrow C_2H_4$ reaction and the planar surfaces separating the two.

present reaction, an infinite potential is assumed along the angle whose constituent planes are perpendicular to the molecular plane of the vinyl radical. The edge of the angle passes through the carbon atom with a vacant orbital and the angle value is 162° . The orientation of the angle is such that one plane of the angle contains the C–C bond and the other is approximately parallel to the C–H bond, cf. Fig. 9. As a result, the interior part of the angle contains the backside potential minimum and the exterior part contains the frontside potential minimum.

Then the individual channel fluxes may be simply evaluated for the portions of configuration space on either side of the potential barrier. Such an assumption removes the overestimate due to the inability to completely separate the two regions with spherical dividing surfaces and also avoids the double counting of flux in the overlap region. This approach does not take into account the flux through the part of the dividing surface which belongs to the angle separating the frontside and backside channel, cf. Fig. 9. This part of the flux can be calculated using Eqs. (2.53) and (2.56). This part of the flux was found to constitute no more than 2% of the overall flux in the temperature range from 200 to 2000 K and, therefore, can be safely neglected. The corresponding predictions for the frontside reaction rate, shown in Fig. 10, are in considerably improved agreement with the trajectory results, being about 10% greater on average. Analogous improvements are found for the backside channel.

The behavior of the microcanonical TS as a function of the energy E and total angular momentum J , follows some trends which seem to be common to different reactive systems. The dependence of the reactive flux on the value of the reaction coordinate is illustrated in Fig. 11 for a range of J values, all at $E = 0.6$ kcal/mol. For a given configuration an increase in J leads to an increase in the rotational kinetic energy $K^{(J)}$, Eq. (2.24), and, as a result, to a decrease, in the internal energy $E - K^{(J)}$, which together with the potential energy $V(Q)$ determines the number of available states, Eq. (2.39). It is well known that the TS expands with decreasing energy due to the increasing importance of enthalpic over entropic effects.²⁵ As a result, just as with a decrease in E , the TS expands with increasing J .

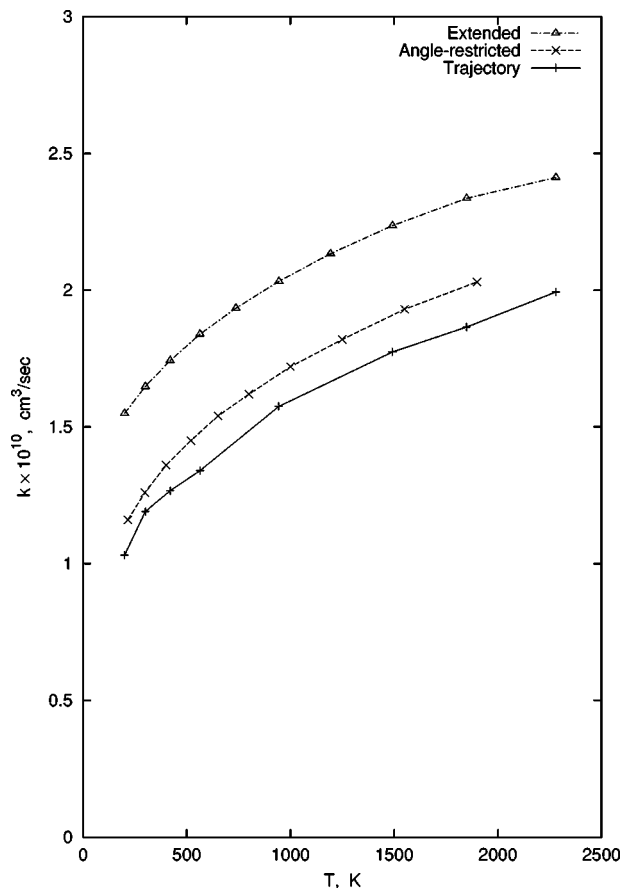


FIG. 10. The frontside rate constant for the reaction $C_2H_3 + H \rightarrow C_2H_4$ is shown as a function of temperature. The results are presented for the E, J -resolved VRC-TST optimizations with the extended set of dividing surfaces, for the E, J -resolved VRC-TST optimizations with an assumed infinite barrier separating the two channels (see the text) and for trajectory simulations.

Interestingly, at the highest J values there is a double minimum in the reactive flux as a function of s . As suggested previously,²⁵ the inner minimum (i.e., that at smaller s values) arises at separations where the covalent interaction is beginning to be significant. The precise location of the inner minimum is due to a balance between the decreasing entropy for the relative and overall bending motions with the decreasing enthalpy for decreasing separations along the minimum energy path. Meanwhile, the outer minimum is related to centrifugal barriers, which arise even for an isotropic potential. These centrifugal barriers increase in size and significance with increasing J values, ultimately producing the second minimum.

IV. CONCLUDING REMARKS

A relatively straightforward and comprehensible derivation of VRC-TST expressions for the canonical, microcanonical, and E, J -resolved reactive flux has been presented. The simplicity of the derivation allows for straightforward generalization to other dividing surfaces, as illustrated by the derivation of analogous results for elliptical and for planar dividing surfaces. The expression employed for the reaction coordinate velocity ultimately yields a more physically illuminating expression for the kinematic factor. This expression

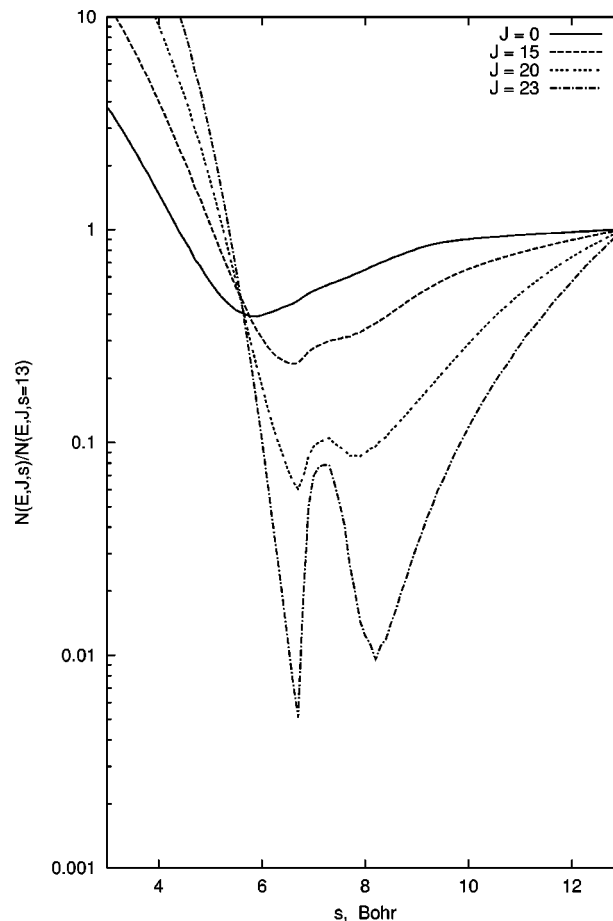


FIG. 11. The E, J -resolved reactive flux normalized to its value at large separation of the pivot points, $s=13$ bohr, is shown as a function of the distance between the hydrogen atom and the pivot point ($x=0, y=2, z=1$) of the vinyl radical at $E=0.6$ kcal/mol and different values of J (in atomic units).

clearly indicates that the kinematic factor increases with increasing deviation of the pivot points from their centers of mass. Furthermore, the evaluation of the present expression for the kinematic factor is straightforward, requiring neither numerical differentiation nor complex analytical transformations.

The application of this methodology to the reaction of $C_2H_3 + H$ has provided a useful confirmation of the accuracy of the VRC-TST approach via comparison with trajectory simulations of the capture process. This application also illustrated some of the difficulties that arise when trying to obtain channel specific estimates. For the frontside channel, channel specific canonical estimates are in error by orders of magnitude at low T , while E, J -resolved results are as much as 50% greater than trajectory results. These errors arise from the difficulty of obtaining channel-specific dividing surfaces which adequately separate the two regions of potential energy surface minima. Such problems may be quite general. Interestingly, the incorporation of approximate dividing surfaces, which at least in principle negate the variational principle, yields channel specific results that are in satisfactory agreement with the trajectory results (i.e., within 10%).

ACKNOWLEDGMENT

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APPENDIX: SOLID ANGLE IN QUATERNION REPRESENTATION

In this section we will show that for a nonlinear molecule the integration over the three-dimensional solid angle $d\Omega$, which in Euler angles can be written as $\sin\vartheta d\vartheta d\varphi d\psi$, is equivalent to the integration over the surface of the four-dimensional sphere in quaternion coordinates. To this end it is convenient to introduce an auxiliary coordinate, R , which is the radius of the vector in the four-dimensional quaternion space,

$$\begin{aligned}\chi &= R \cos(\vartheta/2) \cos(\varphi + \psi)/2, \\ \eta &= R \sin(\vartheta/2) \cos(\varphi - \psi)/2, \\ \xi &= R \sin(\vartheta/2) \sin(\varphi - \psi)/2, \\ \zeta &= R \cos(\vartheta/2) \sin(\varphi + \psi)/2.\end{aligned}\quad (\text{A1})$$

Then the quaternion coordinates are no longer restricted to the unit sphere. Next, we introduce an intermediate transformation defined as

$$u_1 = (\varphi + \psi)/2, \quad u_2 = (\varphi - \psi)/2, \quad (\text{A2})$$

$$v_1 = R \cos(\vartheta/2), \quad v_2 = R \sin(\vartheta/2). \quad (\text{A3})$$

The Jacobians of the transformations given by Eqs. (A2) and (A3) are

$$\frac{\partial(u_1, u_2)}{\partial(\varphi, \psi)} = \frac{1}{2}, \quad \frac{\partial(v_1, v_2)}{\partial(R, \vartheta)} = 2R, \quad (\text{A4})$$

and so the total Jacobian of the transformation from $R, \vartheta, \varphi, \psi$ to u_1, u_2, v_1, v_2 is equal to

$$\frac{\partial(v_1, v_2, u_1, u_2)}{\partial(R, \vartheta, \varphi, \psi)} = R. \quad (\text{A5})$$

The quaternion components χ, η, ξ, ζ can be written in terms of u_i and v_i , as

$$\begin{aligned}\chi &= v_1 \cos u_1, & \zeta &= v_1 \sin u_1, \\ \eta &= v_2 \cos u_2, & \xi &= v_2 \sin u_2.\end{aligned}\quad (\text{A6})$$

From Eq. (A6) one can see that the Jacobians of the transformations from v_1, u_1 to χ, ζ and from v_2, u_2 to η, ξ are

$$\frac{\partial(\chi, \zeta)}{\partial(v_1, u_1)} = v_1, \quad \frac{\partial(\eta, \xi)}{\partial(v_2, u_2)} = v_2, \quad (\text{A7})$$

and so the total Jacobian of the transformation from v_1, v_2, u_1, u_2 to χ, η, ξ, ζ is equal to

$$\frac{\partial(\chi, \eta, \xi, \zeta)}{\partial(v_1, v_2, u_1, u_2)} = v_1 v_2 = \frac{1}{2} R^2 \sin \vartheta. \quad (\text{A8})$$

Multiplying Eq. (A5) by Eq. (A8) one finds that the Jacobian of the transformation from $R, \vartheta, \varphi, \psi$ to χ, η, ξ, ζ , is equal to

$$\frac{\partial(\chi, \eta, \xi, \zeta)}{\partial(R, \vartheta, \varphi, \psi)} = \frac{1}{2} R^3 \sin \vartheta, \quad (\text{A9})$$

which, up to the factor of $1/2$, coincides with the factor of $\sin \vartheta$ in the integral over the solid angle in the Euler angles representation at the fixed $R = 1$.

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¹⁴Equation (2.4), while being similar to the Eq. (72b) of Ref. 10, differs in that the averaging over orbital degrees of freedom in Eq. (2.4) is performed over the direction of the vector $\mathbf{s}$ connecting the pivot points rather than over the direction of the vector $\mathbf{r}$ connecting the fragments' centers of mass, which considerably simplifies the application of the Monte Carlo sampling method to it.

¹⁵Equation (92b) of Ref. 10.

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²²All distances are measured in bohr, the center of mass of the vinyl fragment is at the origin of the reference frame, whose axes are directed along the principal axes of the vinyl fragment, and the orientation of the axes corresponds qualitatively to the one in Fig. 2.

²³See EPAPS Document No. E-JCPSA6-118-011308 for the figures of the optimized radius as a function of the pivot point position at $T = 400$ K for the frontside, backside, and both-sides reaction channels. A direct link to this document may be found in the online article's HTML reference section. The document may also be reached via the EPAPS homepage (<http://www.aip.org/pubservs/epaps.html>) or from <ftp.aip.org> in the directory /epaps/. See the EPAPS homepage for more information.

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